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Contents

0.1 Problemstellung

As AI is more and more incorporated in everydays life raising concers come up about the trustworthiness of these new technologies. Large Language Models like ChatGPT **QuelleTODO**, Llama and many more continuity produce text of high quality and grammatically indistinguishable to human ones. Such aspects seduce the user to believe the generate Texts must be factually correct and spreading of misinformation becomes more frequent. Motivated by these risks, recent work emphasizes the need for automated detection and evaluation methods, especially in scenarios where human verification is infeasible (e.g., cross-lingual use cases where users cannot reliably assess translation correctness). Experts often generalize the phenomenom of LLMs generating illogical or factually incorrect Content as "hallucination".

0.1.1 detection instead of mitigation

An increasing part of recent research focuses on hallucination mitigation, meaning that at the time of generation the output is being checked for hallucination. However, in many practical cases this is not enough, due to inherent problems such as biases, knowledge gaps, and architecture which are fundamentally embedded in the model (Ji et al. 2023, S. 42:19). Hallucination detection is therefore an essential counterpart to hallucination mitigation as it is essential to guarantee post hoc verification and improving the trustworthiness of the systems.

0.1.2 Seed Paper

The seed paper "Can LLMs Detect Intrinsic Hallucination in Paraphrasing and Machine Translation" (Gogoulou et al. 2025) takes this as its starting point. It investigates the inherent ability of different Large Language Models (LLMs) to detect intrinsic hallucination in the two scenarios, paraphrasing (PG) and machine translation (MT). The paper frames intrinsic hallucination as an entailment-based reasoning task and gives a comparison between LLMs and natural language infernce (NLI) models. Henceforth do the authors examine how model size, instruction-tuning, prompt language and prompt formulation impact the results.

0.1.3 My Thesis

Building upon this work, the central thesis of this paper is to examine *how well LLMs are suited for detecting intrinsic hallucinations and which model characteristics are required to perform this task effectively.* Furthermore, this paper discusses different approaches to intrinsic hallucination detection and briefly draws comparisons to its counterpart, extrinsic hallucination.

0.2 Definitions and theoretical basis

0.2.1 Hallucination

In the real world, hallucination is defined as *the perception of an entity or event that is absent in reality* (Macpherson and Platchias 2013, S.TOO). In the context of Language generation, hallucination refers to a models output that on first sight seem fluent and plausible yet is not supported by a given source or context. Ji et al. 2023 defines hallucination as *generated content that is either nonsensical or unfaithful to the provided source content.* In general does the specific definition of hallucination vary between studies yet this a common characterization.

0.2.2 Intrinsic vs. Extrinsic Hallucination

In the context of LLMs, hallucination is commonly further divided into intrinsic and extrinsic hallucination. **Intrinsic hallucination** generally describes cases where the output is deficient with respect to a particular input and or contradicts the input. Additionally, can the deficiency be detected solely by comparing the input and output, without requiring external world knowledge (Ji et al. 2023, S.4). **Extrinsic hallucination** on the other hand, describes cases, that involve information that is not present in the input and cannot be verified solely based on the source alone. In order to detect extrinsic hallucination it typically requires an external knowledge source. It therefore does not matter if the statement is factually correct or not.
Due to these fundamental differences it is necessary to differentiate and develop different methods and metrics. While these categorizations are the broadly shared, the definitions vary from task to task. For example, in abstractive summarization the source is considered the input that the model is supposed to summarize, whereas in a text-to-data the source is considered to be the knowledge-base of the model. Additionally, can the tolerance, what is considered hallucination, change from task to task.(Ji et al. 2023, S.5). Tolerance for hallucinations is very low in summarization and data-to-text tasks because faithfulness to the source is crucial. However, dialogue systems allow relatively higher tolerance, since user engagement and conversational quality are also important goals, especially in open-domain settings.

0.2.3 Faithfulness, Factuality and Entailment

Terms also often used in the context of hallucination are the factuality, faithfulness and entailment. Faithfulness describes how well a model's generated output is consistent by the given input or user provided context. Therefore faithfulness is closely related to intrinsic hallucination. (Ji et al. 2023) describes it as *staying consistent and truthful to the provided source - antonym to "hallucination"*. Thereby if we try to maximize faithfulness of an output it should contain no intrinsic hallucination. As intrinsic hallucination does not require any outside knowledge, faithfulness-oriented evaluation methods or other interference models are particular well suited for this detection task. Especially in conditional generation tasks such as summarization, paraphrasing or machine translation.

On the contrary *Factuality* refers to the quality of being actual or based on fact. However does the interpretation of the "fact" vary. Papers like (Dong et al. 2020) treat the source itself as the factual reference, effectively given the term "factuality" and "faithfulness" the same meaning. For both the seed paper and this thesis, we adopt the definition proposed by Maynez et al. 2020, which clearly distinguishes source knowledge from world knowledge.

0.2.4 Methodische Einordnung

Due to this fundamental difference, intrinsic hallucinations are methodologically easier to evaluate than extrinsic hallucinations. Their detection relies on reasoning about textual consistency rather than knowledge verification, which makes faithfulness-oriented evaluation methods especially suitable. This is particularly relevant for conditional generation tasks such as summarization, paraphrasing, and machine translation, where the source input serves as a clear reference. Building on this observation, the following section focuses on intrinsic hallucination detection and examines how this problem is addressed in the seed paper Can LLMs Detect Intrinsic Hallucinations in Paraphrasing and Machine Translation?

0.3 Seed Paper at the Core: Detecting Intrinsic Hallucinations

This section presents the seed paper "Can LLMs Detect Intrinsic Hallucination in Paraphrasing and Machine Translation?" as the core of this seminar paper. The paper focuses on intrinsic hallucination and explores whether LLMs can reliably detect such errors.

0.3.1 Methods of paper

The seed paper formulates intrinsic hallucination as an entailment problem. Therefore is an output faithful if it is fully entailed by the source input and considered to contain hallucination if it contains additional or contradictory information not logically supported by the source. This formulation allows the detection task to be treated as a reasoning task rather than a knowledge verification problem. To conduct this approach, the authors construct a contrastive evaluation setup. Each instance consists of a source sentence and two candidate hypotheses, where always one is faithful and the other contains intrinsic hallucination. The model is then asked to determine which of the hypothesis is the hallucinated one. This setup ensures a controlled environment and allows for direct comparison across models. The methodology is based on the HalluciGen dataset **QuelleTODO**, which this paper is based upon and specially designed for intrinsic hallucination detection. The dataset convers two szenarios: paraphrasing (PG) and machine translation (MT). For the paraphrasing task, both source and hypotheses are in the same language while the translation task involves cross-lingual input and output. Additionally did the paper test different approaches to prompt the LLM. The study first analyzes the effect of prompt language, testing whether alignment between the prompt language and the language of the source and hypotheses influences performance. In addition, it systematically compares different prompting styles to

evaluate how prompt formulation shapes detection accuracy. Prompts (1-3) contain the term "hallucination" while prompts (3-6) only contain a explicit definition of the concept. Prompt (4-6) on the other hand contain neither and uses formulations that describe hallucination. For example, Prompt 4 is formulated as: *"Given a source sentence (src) and two <scenario> hypotheses (hyp1 and hyp2), detect which one of the two logically contradicts the src."* (Gogoulou et al. 2025, S. 5). Lastly to optimize outputs and ensure proper formatting in addition to the prompts every model receives the same instructions. Models are instructed to only answer with either "hyp1" or "hyp2". **QuelleTODO**modelle die getestet wurden

Versuchsaufbau$_Komplett$

0.3.2 Results of the Paper

SCHREIBE KEYFINDDINGGS VON DEM PAPER

0.4 Alternative Approaches to Instric Hallucination detection

Although the seed paper introduces a specific investigation into the detection of intrinsic hallucination using LLMs and NLI models, it represents only a part of a broader area of research. To better classify these results this section will review alternative approaches and compare them with the entailment-based framework adopted in this seed paper.

0.4.1 Entailment-Based Approaches

The core assumption of intrinsic hallucination is that faithful outputs must be entailed by their source input. Other studies also showed promising results of the capabilities of NLI models for such tasks. Henceforth, has Natural Language Inference (NLI) emerged as a particular fitting system for detecting such inconsistencies. Early studies, such as Maynez et al. 2020, showed that pretrained NLI models can reliably identify inconsistencies in abstractive summarization.

This entailment-based framing is a central point in the seed paper Gogoulou et al. 2025, which explicitly describes intrinsic hallucination detection as a controlled NLI problem. The findings therefore reinforce the view that intrinsic hallucination detection is fundamentally a semantic consistency problem. Therefore are models, that are capable of entailment reasoning a promising solution. The good results of NLIs come with no suprise as the must “handle phenomena like lexical entailment, quan- tification, coreference, tense, belief, modality, and lexical and syntactic ambiguity” (Williams et al., 2018) to successfully predict entailment, contradiction, and neutral relations between sentence pairs. However, **mishra2021react** highlight limitatiations of NLI-based hallucination detection. Standard NLI models are typically trained on sentence-pair datasets. Although Mishra et al. (2021) conduct their empirical analysis in the context of abstractive summarization, the identified limitation of NLI-based evaluation—namely the granularity mismatch between sentence-level entailment training and document-level generation tasks—extends beyond summarization and affects other conditional generation settings as well.

Beyond the entailment-based formulation used in the seed paper using NLIs, other research proposes different approaches for hallucination detection. The survey (**Alansari2025LLMHallucina** formulates different categories for detection methods. While the proposed methods were not originally made for intrinsic hallucination detection, several suit its definiton and are relevant when hallucination is framed as a violation of faithfulness.

0.4.2 Self-Consistency and Uncertainty-based Methods

Another approach is self-consistency-based-detection. For example, (**Manakul2023SelfCheckGPT**) proposes SelfCheckGPT, a framework combining LLM prompting, token-level probability signals and NLI-based entailment checks to detect hallucination on a sentence-level. The main idea of this approach is to generate multiple reponses for the same input evaluate the internal agreement. If the outputs of the model contradict another or vary at certain parts, this instability may imply hallucination. Although the paper does not explicitly differentiate between intrinsic and extrinsic hallucination, its reliance on entailment checks between source and output and the avoidance of external knowledge comparisons makes it particularly applicable for intrinsic inconsistencies. This work is closley related to uncertainty-based and LLM-based self-evaluation approaches. These methods assume that hallucinated content occurs from higher predictive uncertainty. These uncertainties can be then measured through entropy, token-level log probabilites or variance across multiple generations. Nevertheless does this approach has its flaws as it deduces that high confidence collelates with faithfulness. This may lead to cases where small logical errors like negation shifts, incorrect entities or numerical errors stay undected due to the high confidence of the model.

Semantic-based uncertainty methods aim to detect hallucinations by modeling uncertainty at the level of meaning rather than relying solely on token probabilities. Farquhar et al. [89] introduce semantic entropy as a measure for hallucination detection. Their approach generates multiple responses to the same prompt and clusters these outputs based on semantic similarity. Entropy is then calculated across the resulting semantic clusters. High semantic entropy indicates that the model's responses differ substantially in meaning, suggesting a higher likelihood of hallucination. Conversely, low semantic entropy reflects consistent semantic content across generations and therefore greater reliability.

0.4.3 Embedding-based Detection

Embedding-based hallucination detectoin relies on measuring the semantic similiarity between output and the models input or source reference. The underlying assumption is that a faithful output should remain semantically close to its source. If the model generated output differs in the embedding space it could mean a semantic shift happend and therefore it contains hallucination. These approaches are typically further divided into seperate categories. **Similarity-based methods** compares sentences or embeddings of the source and output. **Dale et al.** for example used cross-lingual embedding models such as LaBSE to detect hallucination in machine translation. **Gradient-based Methods** additionally incorparates the gradient of information measuring how sensentive outputs depend on the input. For example, (**Hu2024Embedding**, S.1954) utilize embedding shifts with uncertainty signals to

further imporove detection performance. Overall, embedding-based methods are useful for detecting semantic discrepancies which can be useful for Paraphrasing or Machine Translation tasks.

0.4.4 Prompt-based Approaches

Prompt-based hallucination detection utilizes the exceptional instruction-following capabilities of LLMs by specifically prompting them to evaluate the faithfulness of generated texts. Instead of solely using uncertainty signales or similarty scores, recent studies, like the seed-paper (Gogoulou et al. 2025, used LLMs as the evaluation instance. One key advantage of this approach is the flexiblity. Using LLMs requires no addional training and can generalize across several tasks and languages. Gogoulou et al. 2025 showed that in a contrastive setting LLMs have the inherent ability to determine a hallucinated outcome. While this scenario may archieve decent results in the experiment using LLMs-as-the-Judge also brings its flaws. As shown does the effectivness heavily depend on the model, prompt formulation and the degree of instruction tuning. Additionally, the optimized experimental setup simplifies the generall task by given the LLM a choice between two possible hypothesis. This contrastive framing reduces TODO GUCKEN OB AUSFÜHREN BZGL. LIMITATIONEN

evaluating human language than when assessing machine-generated text, both for categorical and graded annotations. This result aligns with the find- ings by Xu et al. (2024), suggesting that LLMs display a bias towards their own generation. More broadly, this trend calls for caution when using LLMs to automatically evaluate the output of NLP systems.

0.4.5 Important features detection hallucination / conclusion

Taken together, these alternative approaches demonstrate that intrinsic hallucination detection can be addressed through lexical alignment, semantic entailment modeling, probabilistic confidence estimation, or meta-evaluation via prompting. However, each method captures different dimensions of inconsistency and exhibits distinct limitations. This reinforces the insight that intrinsic hallucination detection is fundamentally a problem of semantic consistency rather than merely confidence estimation, and it highlights why entailment-based frameworks, such as the one employed in the seed paper, provide a particularly principled foundation for analyzing model behavior.

0.5 Discussion of usefulness of LLMS

Building upon the idea of using LLMs as a detection method for intrinsisch hallucionation, a more fundamental question arises: *are LLMs inherently suited to detect hallucination?* This question is closely related to the big research field of LLM-as-a-Judge.

0.5.1 LLM-as-Judge Perspective

0.5.2 Non trivial Annotaiton

It is important to note that detecting hallucination is not even trivial for humans. The JUDGE-BENCH (**tan2024judgebench** study shows varying agreement not only between humans and models, but also among human annotator themself. The reports in Table 3 (pp.11-12), show that inter-annotator agreement (e.g. Krippendorff's alpha between humans vary strongly for certain tasks, indicating that even humans frequently disagree on properties like coherence, consistency and faithfulness. These results suggest that intrinsic hallucination may not be a binary or objectively trivial problem but often requires interpretation. Especially in cases of faithfulness and entailment it is not trivial to determine whether a text should be considered to contain hallucination or not.

Another important finding from the JUDGE-BENCH Paper is that LLM judges correlate more strongly with non-expert annotations rather that with expert judgments (**Bavaresco2025JudgeBench** S. 241). The authors offer an unsupported interpretation that non-experts may rely more on surface level features, whereas expert judges apply stricter criteria. Believing this thesis this discrepancy could raise concerns whether LLMs truly capture faithfulness or rather approximate surface believablity. However this claim requires more studying.

0.5.3 Systemic biasis of LLMS

A deeper theoretical limitation emerges from the survey's acknowledgment that LLM-as-a-Judge systems are themselves generative models subject to hallucination . Since an LLM judge relies on its internal knowledge representation, it may incorrectly classify a true statement as false due to outdated or incomplete knowledge, or conversely fail to detect a fabricated claim if it appears plausible within its learned distribution. This recursive vulnerability implies that LLMs cannot serve as infallible arbiters of truth; they remain probabilistic approximators of human-like reasoning.

0.5.4 Theoretical Limits

0.5.5 Conlusion LLMs for detection

Taken together, the findings of the survey suggest a nuanced conclusion. LLMs are potentially suitable for hallucination detection because they combine semantic reasoning with scalability, outperforming shallow automated metrics . However, their suitability is conditional rather than absolute. Reliability depends on validated agreement with human experts, systematic bias mitigation, adversarial robustness testing, careful prompt engineering, and possibly ensemble integration. Moreover, because LLM judges can themselves hallucinate, they cannot guarantee objective truth verification.

Therefore, based the survey's framework, LLMs should not be viewed as autonomous truth detectors but as probabilistic evaluators whose effectiveness in hallucination detection depends on rigorous reliability engineering. They are suitable under controlled, validated conditions, but unsuitable as standalone, uncalibrated arbiters of factual correctness.

0.6 General quesiotn

0.7 Outlook seedpaper

0.7.1 Zukunftsperspektive

Building on the findings of the seedpaper (Gogoulou et al. 2025), a next step could be to extend the current annotation framework towards a more scalable, model-assisted data creation. The paper already partially relies on artificial annotations, as intrinsic hallucinations are manually created through controlled perturbations such as negations, entity substitutions etc.

0.7.2 Conclusion

Due to the broadness of tasks with which LLMs are often used for, there are several domains which have utilized unique strategies for assessing faithfulness, with these methodologies having been specifically chosen and optimised for the domain at hand. Faithfulness is a crucial evaluation criterion for all domains within the scope of this review, as users rely on the model to provide information that is factually correct and true to the original source otherwise problems will arise with varying levels of severity depending on the criticality of the task at hand. It can be challenging to define faithfulness universally due to the intricacies required across domains. For example, summarization requires the output to be aligned with the source material, which can be validated without the requirement of a ground truth. Whereas an open question-answering (QA) task requires a ground truth to be provided for the faithfulness to be measured.

A Zusätzliche Materialien

Bibliography

Dong, Yue et al. (2020). "Multi-Fact Correction in Abstractive Text Summarization". In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, pp. 9320–9331.

Gogoulou, Evangelia, Shorouq Zahra, Liane Guillou, Luise Dürlich, and Joakim Nivre (2025). "Can LLMs Detect Intrinsic Hallucinations in Paraphrasing and Machine Translation?" In: *arXiv preprint arXiv:2504.20699*. arXiv: 2504.20699 [cs.CL]. URL: https://arxiv.org/abs/2504.20699.

Ji, Ziwei et al. (2023). "Survey of Hallucination in Natural Language Generation". In: *ACM Computing Surveys* 55.12. DOI: 10.1145/3571730. URL: https://doi.org/10.1145/3571730.

Macpherson, Fiona and Dimitris Platchias (2013). *Hallucination: Philosophy and Psychology*. Cambridge, MA: MIT Press.

Maynez, Joshua, Shashi Narayan, Bernd Bohnet, and Ryan McDonald (2020). "On Faithfulness and Factuality in Abstractive Summarization". In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Online: Association for Computational Linguistics, pp. 1906–1919. DOI: 10.18653/v1/2020.acl-main.173.